Standard Deviation & Variance

Exam code: 9MA0

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Standard deviation & variance

- The **variance** is another measure for the spread of the data, it measures the variability from the mean of the data

What is the variance and the standard deviation?

- The variance is a statistic that tells us how **varied** a set of data is
  - Data that is more spread out will have a greater variance
  - Data that is consistent and close together will have a smaller variance
- The **standard deviation** is the square root of the variance
- The symbol for the population standard deviation is the lowercase Greek letter sigma, σ and for variance is sigma squared, σ^2
- Standard deviation and variance are used interchangeably within this course so make sure you look out for which one a question shows or asks for

How are the variance and standard deviation calculated?

- There is more than one formula that can be used for calculating the variance, and you should choose the most

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

divided by the frequency

- Variance = $\frac{\Sigma(x-\bar{x})^2}{n}$
  - This formula can be time consuming and therefore is rarely used in this statistics course

- A second, **easier to use version of the variance** is:
  - Variance = $\frac{\Sigma x^2}{n} - \bar{x}^2$
  - This version is easier to work with and should be used in most instances
    - The **summary statistic** $S_{xx} = \Sigma(x_i - \bar{x})^2 = \Sigma x_i^2 - \frac{(\Sigma x_i)^2}{n}$ can help derive different formulae as shown below:

Variance =

$$\sigma^2 = \frac{\Sigma(x-\bar{x})^2}{n} = \frac{S_{xx}}{n} = \frac{1}{n}\left(\Sigma x^2 - \frac{(\Sigma x)^2}{n}\right) = \frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2 = \frac{\Sigma x^2}{n} - \bar{x}^2$$

- An easy way to remember this is to think of it as '*the mean of the squared values minus the square of the mean*'
  - Most calculators can be used to find summary statistic such as the standard deviation and variance fairly quickly, practice finding it on yours

- The **standard deviation** is the square root of the variance
  - Standard deviation =

    $$\sigma = \sqrt{\frac{\Sigma(x-\bar{x})^2}{n}} = \sqrt{\frac{\Sigma x^2}{n} - \bar{x}^2} = \sqrt{\frac{S_{xx}}{n}}$$

    - Make sure you know how to find these formulae in the formula booklet and are familiar with the version given

- The units for standard deviation are the same as the units for the data and the units for variance are the same as the units for the data but squared

How are the variance and standard deviation calculated from a frequency table?

- The method for finding the variance from a frequency table is similar to that of the mean
  - If calculating from a grouped frequency table, find the midpoints, x, first
  - Multiply each x value by its corresponding frequency and use these values within the formulae
  - The formulae will become
    - Variance = $\sigma^2 = \frac{\Sigma f(x-\bar{x})^2}{\Sigma f} = \frac{\Sigma f x^2}{\Sigma f} - \left(\frac{\Sigma f x}{\Sigma f}\right)^2$
    - Standard deviation =

      $$\sigma = \sqrt{\frac{\Sigma f(x-\bar{x})^2}{\Sigma f}} = \sqrt{\frac{\Sigma f x^2}{\Sigma f} - \left(\frac{\Sigma f x}{\Sigma f}\right)^2}$$

Worked Example

Time (minutes)	Frequency
$2 \le t < 4$	1
$4 \le t < 6$	4
$6 \le t < 8$	7
$8 \le t < 10$	5
$10 \le t < 12$	2
$12 \le t < 20$	2

(i) Calculate an estimate of the variance of the time taken to complete a question, include units with your answer.

(ii) Write down an estimate of the standard deviation of the time taken to complete a question, include units with your answer.

Answer:

(i)

Find the midpoint, h, of each group:

Time, (mins)	Midpoint, t	t^2	Frequency, f	ft	ft^2
$2 \leqslant t < 4$	3	9	1	3	9
$4 \leqslant t < 6$	5	25	4	20	100
$6 \leqslant t < 8$	7	49	7	49	343
$8 \leqslant t < 10$	9	81	5	45	405
$10 \leqslant t < 12$	11	121	2	22	242
$12 \leqslant t < 20$	16	256	2	32	512

This group has a different width to the others.

Find the summary statistics:

Your calculator can be used in 'STAT' mode to find these values

$\Sigma ft = 171 \quad \Sigma ft^2 = 1611 \quad \Sigma f = 21$

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(i) Substitute into the formula for variance

$$\sigma^2 = \frac{\Sigma ft^2}{\Sigma f} - \left(\frac{\Sigma ft}{\Sigma f}\right)^2 = \frac{1611}{21} - \left(\frac{171}{21}\right)^2$$

$$= \frac{510}{49} = 10.408$$

Estimated variance = 10.4 minutes^2 (3.s.f.)

(ii) The standard deviation is the square root of the variance

$$\sigma = \sqrt{\frac{510}{49}} = \frac{\sqrt{510}}{7} = 3.2261...$$

Estimated standard deviation = 3.23 minutes (3.s.f.)

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Examiner Tips and Tricks

- Look out for whether a question gives or asks for the standard deviation or variance, especially if the question

formulae will be possible. On the rare occasion that you are asked to calculate directly from a table think carefully about which version of the formula is quickest and easiest to use. It will almost always be the second version given in this revision note.

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